SCOTT FOXLEY-BERRY

Backend & Platform Engineer

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SUMMARY

Backend and platform engineer with 8 years of experience building and operating production systems. Design Go services and automation on Kubernetes and GKE at The Home Depot for a platform serving more than 4,000 internal services and over 2 billion requests a day, and lead incident response for it. Own work end to end, from requirements through deployment, observability, and operational support, and apply a data-science background (M.S., Indiana University) to reliability and analysis problems.

TECHNICAL SKILLS

Languages: Go, Python, SQL, Bash; also Rust, Scala

Backend: REST APIs, OpenAPI, PostgreSQL, GraphQL clients, server-rendered web (htmx), distributed systems

Platform & Cloud: Kubernetes, GKE, Docker, Terraform, Ansible, AWS, Linux, Railway

Reliability & CI/CD: GitHub Actions, Prometheus, Grafana, incident response and root-cause analysis, automated testing, release automation

Data: BigQuery, Kafka, Elasticsearch, MongoDB, Spark, ETL pipelines, Polars, NumPy, Numba, statsmodels, NLP, machine learning

PROFESSIONAL EXPERIENCE

The Home Depot — Systems Engineer II

Dec 2020 – Present

Atlanta, Georgia

- Design and build backend services, APIs, and operational tooling in Go and Python for a Kubernetes/GKE platform serving more than 4,000 internal services and over 2 billion requests a day
- Lead incident response, root-cause analysis, and long-term remediation for critical production issues and service disruptions
- Own CI/CD pipelines in GitHub Actions that automate testing, validation, and deployment across Go and Python projects
- Build observability with Prometheus, Grafana, and custom telemetry to speed up debugging and improve service reliability
- Turn requirements from infrastructure, security, and application teams into scalable self-service tooling

Selected projects

- **Watchdog:** architected, and continue to lead, a production Go service that makes traffic-advertisement and health-validation decisions for thousands of services. Rule-based and anomaly-detection health evaluation determines service readiness and traffic eligibility. Exposes REST/JSON APIs with OpenAPI specifications, automated API documentation, and Prometheus metrics
- **Bifrost Bot:** designed and maintain a Go Slack application on GKE that automates infrastructure promotions, service health reporting, documentation discovery, and customer escalations, integrating multiple internal systems
- **Infrastructure validation:** created a centralized smoke-testing platform (Go, Python, GitHub Actions) that validates infrastructure health and deployment readiness, with automated certificate-expiration detection, service validation, and documentation quality checks
- **Proxy migration data platform:** led an ETL and migration platform for moving customer services to a new proxy architecture. Python and BigQuery workflows apply NLP and machine-learning classification to identify ownership and migration requirements, reducing manual triage and speeding adoption

Featurespace — Implementation Engineer

Jul 2019 – Dec 2020

Atlanta, Georgia

- Deployed and supported financial fraud-detection platforms on AWS-hosted Linux infrastructure

- Operated Kafka, Elasticsearch, MongoDB, and Zookeeper for large-scale event processing and analytics workloads
- Automated deployment, configuration management, and operational support in Python and Bash
- Supported customers through system integrations, data migrations, and production deployments

Noble Systems Corporation — Implementation Engineer May 2018 – Jul 2019

Atlanta, Georgia

- Installed, configured, and supported enterprise software platforms across Linux and Windows environments
- Administered PostgreSQL databases, including configuration management, operational reporting, and customer data
- Wrote Python and Bash automation for deployment and operations, and delivered technical training to customers

PROJECTS

Magic 8-Ball — live web app for APA pool league teams 2022 – Present

Go, PostgreSQL, htmx, Docker, GitHub Actions, Railway · Live demo: magic-8ball.up.railway.app/demo

- Built and run a production web app, in private beta with real league teams, for attendance, legal-lineup checks, and match-night tools. Rosters, skill levels, and schedules sync from the APA's GraphQL API every 15 minutes
- Hardened it for the public internet: APA sign-in without storing passwords, hashed session tokens, a strict Content Security Policy, same-origin checks on writes, rate limiting, and parameterized SQL
- Made CI the deploy gate: staticcheck, race-enabled tests against Postgres with a coverage floor, govulncheck, Docker builds, and headless-Chrome smoke tests, with separate dev and production environments
- Migrated it from a Python Discord bot to a ~29 MB distroless Go service, with parity tests against the original outputs and 21 architecture decision records

Shelved — daily word puzzle for BookTok readers 2026 – Present

Go, PostgreSQL, JavaScript, Claude Code, Docker, Railway · Live: shelved.up.railway.app

- Launched and run a free daily puzzle game (Go, Postgres, framework-free front end, no player accounts or trackers). A Claude Code skill drafts 30-day puzzle batches, CI validates each batch, and rejecting a day on the dev site reschedules the rest
- Gated every deploy on CI (staticcheck, race tests on Postgres, coverage floor, govulncheck, headless-Chrome smoke tests), with a strict CSP, a dev-only admin proven absent from production by tests, and 20 decision records

Pairs Engine — statistical-arbitrage research platform 2026 – Present

Python, Polars, NumPy, Numba, statsmodels, Parquet, Astro · Results: berryscottr.github.io/berryscottr/quant

- Scan 1,000 to 3,000 US stocks each week for cointegrated pairs using vectorized Engle-Granger and ADF tests, verified against statsmodels, on resumable, validated minute-bar data ingestion
- Built a walk-forward, minute-bar backtester with realistic costs and no look-ahead. All 4,699 configurations tried are logged for Deflated Sharpe and overfitting-probability checks, and a one-shot holdout was read against pre-registered pass bands
- Engineered for reproducibility: bit-identical results regardless of worker count, provenance fingerprints on every derived artifact, and more than 140 automated tests

EDUCATION

Indiana University — Master of Science in Data Science 2022

Machine learning and cybersecurity concentration; GPA 3.63. Research project on detecting crime rationalizations in online discussions using NLP and classification models.

Georgia Institute of Technology — Bachelor of Science in Industrial Engineering 2018

Quality and statistics concentration. Senior design project with TUG Technologies, an integer-programming platform to improve assembly-line efficiency, won Overall Best Project in Georgia Tech Industrial Engineering.

ACTIVITIES & VOLUNTEERING

Competitive swimmer · Billiards team captain · Former swim team head coach and swimming referee · Math tutor · Trees Atlanta volunteer · Special Olympics fundraiser and donor